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Tutorial ①

① (10) (a) $\sim(p \to q) \to p$

p	q	$p \to q$	$\sim(p \to q)$	full
1	1	1	0	1
1	0	0	1	1
0	1	1	0	1
0	0	1	0	1

(b) $\sim(p \to q) \to \sim q$

p	q	$p \to q$	$\sim(p \to q)$	$\sim q$	full
1	1	1	0	0	1
1	0	0	1	1	1
0	1	1	0	0	1
0	0	1	0	1	1

(9) $p \to q$ if p then q a) true b) true

8) u is odd or -q is positive. (c) u is even or -q is negetive

Ans = a, b

(7) (a) $p \vee q \equiv q \vee p$

p	q	$p \vee q$	q	p	$q \vee p$
1	1	1	1	1	1
1	0	1	1	0	1
0	1	1	0	1	1
0	0	0	0	0	0

True.

(b) $\neg(p \wedge q) \equiv \neg p \vee \neg q$

p	q	$p \wedge q$ (and)	$\neg(p \wedge q)$	$\neg p$	$\neg q$	$\neg p \vee \neg q$ (or)	$\neg p \cap q \equiv \neg p \vee \neg q$
1	1	1	0	0	0	0	1
1	0	0	1	0	1	1	1
0	1	0	1	1	0	1	1
0	0	0	1	1	1	1	1

True

(c) $(p \vee q) \vee r \equiv p \vee (q \vee r)$

p	q	r	$p \vee q$	$(p \vee q) \vee r$	$q \vee r$	$p \vee (q \vee r)$	$\equiv$
1	1	1	1	1	1	1	1
1	1	0	1	1	1	1	1
1	0	1	1	1	1	1	1
1	0	0	1	1	0	1	1
0	1	1	1	1	1	1	1
0	1	0	1	1	1	1	1
0	0	1	0	0	1	1	0
0	0	0	0	0	0	0	1

False.

(6) $p \wedge \neg q$ (b)

5) $q \to p$. if p then q (b) if I am in Bangalore then I love cricket

4) (a) the only odd prime no is 2.

$p \leftrightarrow q \oplus (p \leftrightarrow \neg q)$.

Exclusive or.

p	q	$\neg q$	$p \leftrightarrow \neg q$	$p \leftrightarrow q$	$p \leftrightarrow q \oplus (p \leftrightarrow \neg q)$
1	1	0	0	1	1
1	0	1	1	0	1
0	1	0	1	0	1
0	0	1	0	1	1

(tautology)

2) when I stay up late it is neccesary tat I sleep till noon

(q is neccasry for p) $\sim(p \to q)$

① when I wake up early it's necessary that I don't sleep till noon

② when I don't stay up late it is neccary that I don't sleep till noon

aren't my ans $(\neg p \to \neg q)$ How to write $\sim(p \to q)$?

① $p \to q$ → if p then q q is nec for p

(a) r is necess for p Ans = if you get an A on the final Exam then you get an A in this class.

p ~~suff~~ q.

(b) $p \wedge q$ suff. r.

$(p \wedge q) \to r$ if you get an A on the final exam and do every exercise in the book then you get an a in this class.

(p only if q)

(c) $r \to (q \vee p)$.

if you get an A in this class then you do every exercise in this book or get an A on the final Exam.

done ✓